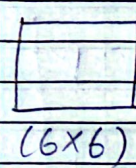


Backpropagation in CNN

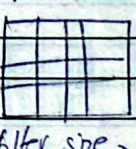
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image



filter



filter size = (3×3)

strides = 1

bias b_1

(filter bias)

$$n + 2p - f + 1$$

$$= 6 - 3 + 1$$

$$= 3 + 1$$

$$= 4$$

$$(4 \times 4)$$

$$(4 \times 4)$$

1) Trainable parameters

conv layer

$$w_1 = (3, 3)$$

$$b_1 = (1, 1)$$

$$w_2 = (1, 4)$$

$$b_2 = (1, 1)$$

$$= (3 \times 3) + 1 + 1 + 4 + 1 + 1$$

$$= 9 + 1 + 4 + 1$$

$$= 15 \text{ trainable parameters}$$

fully connected layers

CNN part

ANN part

$$w_2 = (1 \times 4)$$

$$b_2 = (1, 1)$$

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max

pooling

pool size

$$= (2 \times 2)$$

$$S = 2$$

$$= (2, 2)$$

$$(4 \times 1)$$

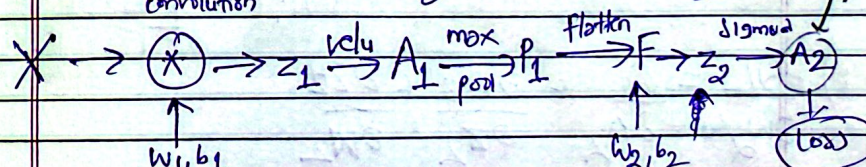
$$(1)$$

$$\text{Loss}$$

$$\text{Binary (cross) Entropy Loss} = -y_i \log(\hat{y}_i) - (1 - y_i) \log(1 - \hat{y}_i)$$

2) forward propagation

Logical Diagram



$$z_1 = \text{conv}(X, w_1) + b_1$$

$$A_1 = \text{relu}(z_1)$$

$$P_1 = \text{maxpool}(A_1)$$

$$F = \text{Flatten}(P_1)$$

$$z_2 = w_2 F + b_2$$

$$A_2 = \sigma(z_2)$$

Gradient Descent

Full Connected
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matrix matrix matrix

$$W_1 = W_1 - \eta \frac{\partial L}{\partial W_1}$$

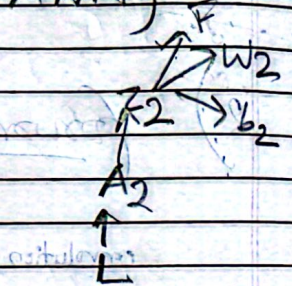
$$W_2 = W_2 - \eta \frac{\partial L}{\partial W_2}$$

$$b_1 = b_1 - \eta \frac{\partial L}{\partial b_1}$$

$$b_2 = b_2 - \eta \frac{\partial L}{\partial b_2}$$

1) Backpropagation of fully connected layer (ANN)

ie. $\frac{\partial L}{\partial W_2} \frac{\partial L}{\partial b_2}$



$$(i) \frac{\partial L}{\partial W_2} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial Z_2} \times \frac{\partial Z_2}{\partial W_2}$$

$$(ii) \frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial Z_2} \times \frac{\partial Z_2}{\partial b_2}$$

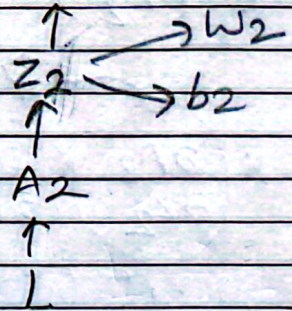
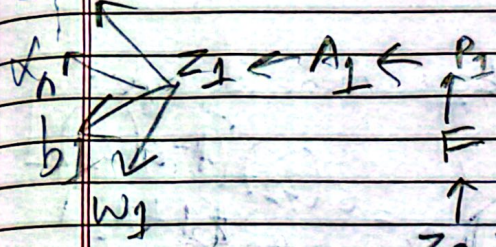
2) Backpropagation of Convolution layer

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$$(iii) \frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial Z_2} \times \frac{\partial Z_2}{\partial F} \times \frac{\partial F}{\partial P_1} \times \frac{\partial P_1}{\partial A_1}$$

$$\times \frac{\partial A_1}{\partial Z_1} \times \frac{\partial Z_1}{\partial W_1}$$

from max pooling



from Flatten from max pooling

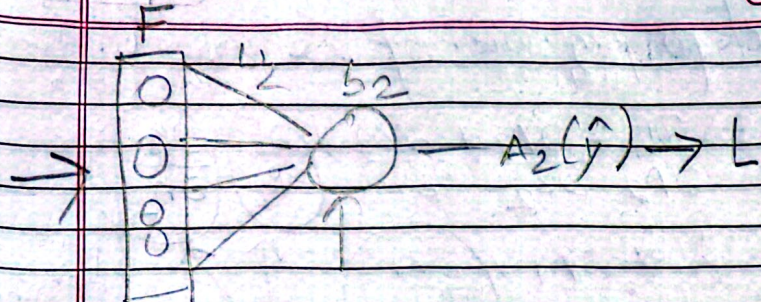
$$(iv) \frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial Z_2} \times \frac{\partial Z_2}{\partial F} \times \frac{\partial F}{\partial P_1} \times \frac{\partial P_1}{\partial A_1}$$

$$\times \frac{\partial A_1}{\partial Z_1} \times \frac{\partial Z_1}{\partial b_1}$$

from max pooling

Fully connected layer part (ANN part)

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forward prop eqf;

$$z_2 = w_2 F + b_2$$

$$A_2 = \sigma(z_2) = \sigma(z_2) [1 - \sigma(z_2)] = a_2 [1 - a_2]$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial z_2} \times \frac{\partial z_2}{\partial w_2} \quad y = F$$

A to z for single nodes

$$\frac{\partial L}{\partial a_2} = \frac{\partial L}{\partial a_2} [-y_i \log(a_2) - (1 - y_i) \log(1 - a_2)]$$

$$= -\frac{y_i}{a_2} + \frac{(1 - y_i)}{1 - a_2}$$

$$= \frac{-y_i(1 - a_2) + a_2(1 - y_i)}{a_2(1 - a_2)}$$

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$$= \frac{-y_i + y_i a_2 + a_2 - a_2 y_i}{a_2(1 - a_2)}$$

$$= \frac{(a_2 - y_i)}{a_2(1 - a_2)}$$

So, $\frac{\partial L}{\partial w_2} = \frac{(a_2 - y_i)}{a_2(1 - a_2)} \times \frac{\partial z_2}{\partial w_2} \times F$

bias

$$= (a_2 - y_i) F$$

for all,

$$(1) \quad \frac{\partial L}{\partial w_2} = \frac{1}{M} (A_2 - Y) F^T$$

for 2 batch,

Transpose

(1x1) (1x1) (1x1)

(1x1) (1x1)

(1,1)

$$(1) \quad \frac{\partial L}{\partial b_2} = \frac{1}{M} (A_2 - Y)$$

for convolution layers (Maxpooling, flatten, Page No.:
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$(A2 - Y)$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial z_2} \times \frac{\partial z_2}{\partial F} \times \frac{\partial F}{\partial P_1} \times \frac{\partial P_1}{\partial A_1} \times \frac{\partial A_1}{\partial w_1}$$

6

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial z_2} \times \frac{\partial z_2}{\partial F} \times \frac{\partial F}{\partial P_1} \times \frac{\partial P_1}{\partial A_1} \times \frac{\partial A_1}{\partial b_1}$$

5

Already
calculated
in
ANN
part

6 derivatives to
find

1) $\frac{\partial z_2}{\partial F} = w_2$

2) $\frac{\partial F}{\partial P_1} = \text{reshape}(P_1 \cdot \text{shape})$ (no trainable parameters, just do opposite of what it was done)

3) $\frac{\partial F}{\partial A_1}$

3)

$$\frac{\partial L}{\partial P_1} = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$$

(2x2)

↙

(4x4)

$$\begin{bmatrix} 0 & x_1 & x_2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & x_3 & x_4 & 0 \end{bmatrix}$$

Reverse of
max
pooling

(no trainable parameters, Page No.:
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in max pooling layer, do just do the opposite of what max pooling was done)

A) $\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0 & 5 \\ 7 & 6 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 12 & 18 \end{bmatrix}$

$\begin{bmatrix} 0 & 5 \\ 7 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 10 & 14 \\ 16 & 15 \end{bmatrix}$

3)

$$\frac{\partial L}{\partial A_1} = \begin{cases} \frac{\partial L}{\partial P_{1xy}} & \text{if } A_{mn} \text{ is the max element} \\ 0 & \text{otherwise} \end{cases}$$

relu

$$4) \frac{\partial A_1}{\partial z_1} = \begin{cases} 1 & \text{if } z_1 > 0 \\ 0 & \text{if } z_1 < 0 \end{cases}$$

$$5) \frac{\partial z_1}{\partial b_1}$$

convolution layer
(has parallel
parameters)

Let's analyze from an example,

$$\begin{matrix} X \rightarrow (*) \rightarrow z_1 \\ (3,3) \quad \uparrow \quad (2,2) \\ \quad \quad w_1/b_1 \end{matrix}$$

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} \quad w_1 = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$$

Filter size

$$z_1 = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$$

$$z_{11} = x_{11}w_{11} + x_{12}w_{12} + x_{21}w_{21} + x_{22}w_{22} + b_1$$

$$z_{12} = x_{12}w_{11} + x_{13}w_{12} + x_{22}w_{21} + x_{23}w_{22} + b_1$$

$$z_{21} = x_{21}w_{11} + x_{22}w_{12} + x_{31}w_{21} + x_{32}w_{22} + b_1$$

$$z_{22} = x_{22}w_{11} + x_{23}w_{12} + x_{32}w_{21} + x_{33}w_{22} + b_1$$

Now,

$$\frac{\partial L}{\partial z_1} = \begin{bmatrix} \frac{\partial L}{\partial z_{11}} & \frac{\partial L}{\partial z_{12}} \\ \frac{\partial L}{\partial z_{21}} & \frac{\partial L}{\partial z_{22}} \end{bmatrix}$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial b_1}$$

$$= \frac{\partial L}{\partial z_{11}} \times \frac{\partial z_{11}}{\partial b_1} + \frac{\partial L}{\partial z_{12}} \times \frac{\partial z_{12}}{\partial b_1} + \frac{\partial L}{\partial z_{21}} \times \frac{\partial z_{21}}{\partial b_1} + \frac{\partial L}{\partial z_{22}} \times \frac{\partial z_{22}}{\partial b_1}$$

$$= \frac{\partial L}{\partial z_{11}} + \frac{\partial L}{\partial z_{12}} + \frac{\partial L}{\partial z_{21}} + \frac{\partial L}{\partial z_{22}} = \text{sum} \left(\frac{\partial L}{\partial z_1} \right)$$

$$(5) \frac{\partial L}{\partial b_1} = \text{sum} \left(\frac{\partial L}{\partial z_1} \right) \text{ similar}$$

Now,

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial w_1}$$

$$\frac{\partial L}{\partial w_1} \left[\begin{array}{c} \frac{\partial L}{\partial w_{11}} \quad \frac{\partial L}{\partial w_{12}} \\ \frac{\partial L}{\partial w_{21}} \quad \frac{\partial L}{\partial w_{22}} \end{array} \right]$$

$$\frac{\partial L}{\partial z_1} \left[\begin{array}{c} \frac{\partial L}{\partial z_{11}} \quad \frac{\partial L}{\partial z_{12}} \\ \frac{\partial L}{\partial z_{21}} \quad \frac{\partial L}{\partial z_{22}} \end{array} \right]$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial z_{11}} \times \frac{\partial z_{11}}{\partial w_{11}} + \frac{\partial L}{\partial z_{12}} \times \frac{\partial z_{12}}{\partial w_{11}} + \frac{\partial L}{\partial z_{21}} \times \frac{\partial z_{21}}{\partial w_{11}} + \frac{\partial L}{\partial z_{22}} \times \frac{\partial z_{22}}{\partial w_{11}}$$

$$+ \frac{\partial L}{\partial z_{22}} \times \frac{\partial z_{22}}{\partial w_{11}}$$

$\frac{\partial L}{\partial w_1}$ = similar format

$$\frac{\partial L}{\partial w_2} = //$$

$$\frac{\partial L}{\partial w_3} //$$

$$\frac{\partial L}{\partial w_4} //$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial z_{11}} \times \frac{\partial z_{11}}{\partial w_{11}} + \frac{\partial L}{\partial z_{12}} \times \frac{\partial z_{12}}{\partial w_{11}} + \frac{\partial L}{\partial z_{21}} \times \frac{\partial z_{21}}{\partial w_{11}} + \frac{\partial L}{\partial z_{22}} \times \frac{\partial z_{22}}{\partial w_{11}}$$

$$+ \frac{\partial L}{\partial z_{22}} \times \frac{\partial z_{22}}{\partial w_{11}}$$

$$\frac{\partial L}{\partial z_{12}}$$

$$\frac{\partial L}{\partial z_{21}}$$

$$\frac{\partial L}{\partial z_{22}}$$

So, we can see pattern,

$$\frac{\partial L}{\partial w_1} = \text{conv} \left(\frac{\partial L}{\partial z_{11}} \right)$$